Decision log: the embodied-fly path and its branch points

*Every crossroads on the way from the chip-to-animal study to the embodied fly, with the options that were open, the one taken, why, and where to branch from to explore another. Results are not restated here: each stage's numbers live in `report/ROADMAP_EMBODIED_FLY.md` §7, cited to `outputs/metrics.json`, and each stage's registration in `preregistration/`. Commits are on master unless marked otherwise. Decisions taken by the user are marked **(user)**; the rest were taken by the agent and are open to revision.*

How to explore another branch

Check out the commit named in the decision, create a branch, register the alternative in `preregistration/` before running it (fresh seeds if the original result has been seen), and run it with its own command or `python -m neuromps.fly.pipeline --only <step>`.

Decisions

D1. Which fly connectome release the model runs on

- **Options:** FlyWire 630 (the published model's release), FlyWire 783 (the current one).
- **Taken:** 783 for the studies, with a registered 630-vs-783 replication of the published sugar result. **Why:** the current release, with the published release kept as a check.
- **Revisit at:** `fly_chip_to_animal` registration; `neuromps.fly.connectome.load_630`.

D2. Which body simulator for the embodied fly

- **Options:** FlyGym / NeuroMechFly v2 (needs Python 3.12), the FlyMimic musculoskeletal leg loaded in plain MuJoCo, flybody.
- **Taken:** FlyMimic in plain MuJoCo (pinned `mujoco` as the optional `embodied` extra); flybody later for the proboscis. **Why:** muscle-driven, runs in the project's Python 3.11, no new framework. **Revisit:** a FlyGym environment would give six muscle-driven legs only if its musculoskeletal model covers them.

D3. After stage 1 failed: what to change first

- **Options:** rescale the male-CNS synaptic weight further; drive the male nerve cord from descending neurons with the brain removed; switch connectome.
- **Taken:** the nerve cord from descending neurons (stage 1c), then the FlyWire brain joined to it through type-matched descending neurons (stages 1e, 1f). **Why:** the male brain avalanches under the published parameters (`fly_spread`); each connectome is used where its model is validated.

D4. Stage 1e's precondition failed on an absolute rate line

- **Taken:** stage 1f, a relative precondition and depth of reach, on fresh seeds; 1e stays uninterpretable. **Why:** the absolute line had no basis.

D5. Stage 2a failed: how to look for a walking rhythm (user: sweep)

- **Options offered:** fix adaptation parameters from literature; sweep them and report the whole sweep. **Taken (user):** the sweep (stage 2c), with a registered selection rule and confirmation.
- **Revisit:** published adaptation values, if any appear for fly leg or premotor neurons.

D6. The glutamate sign rule (user: go ahead)

- **Taken by the agent:** two bounding rules (glutamate excitatory onto motor neurons only; and everywhere), each with the adaptation grid (stage 2d). **Unexplored:** under the corrected rhythm measure the selection rule would pick (glutamate everywhere, 16 mV, 100 ms); it was not confirmed on fresh seeds.

D7. A different neuron model (user: look at how others make flies walk)

- **Found:** the walking flies online use trained controllers (Eon Systems, flybody); only Pugliese et al. (2025) get a rhythm from nerve-cord wiring, with a firing-rate model.
- **Taken:** reimplement their model from the paper and configuration (their code is unlicensed).
- **Size stand-in (stage 2e):** total synapse count, because the male CNS has no volumes. This starved DNg100; stage 2f drove every command neuron to the same rate; stage 2g used their front-leg network and recruitment rule; stage 2h a geometric input search.
- **Unexplored:** neuron volumes (a future male-CNS release, or MANC volumes through the male-CNS-to-MANC matching); their own network files under a licence.

D8. A flaw in the registered rhythm measure

- **Taken:** a registered audit with a peak-requiring measure (prominence 0.1, calibrated on synthetic signals only), re-scoring stages 1d-2e on their original seeds; registered outcomes kept as recorded, the documents carry both. Stage 1d's D2 withdrawn.
- **Revisit:** prominence 0.05 or 0.2 give the same calibration results; a different peak criterion could be registered.

D9. Where to close the loop first (stage 3)

- **Options:** the front leg (muscle-driven, thinly traced proprioceptors), a middle or hind leg (well traced, but only position-servoed in flybody), the proboscis.
- **Taken:** the front leg through the claw neurons; the unknown flexion/extension tuning bracketed by two assignments and a sham; then a perturbation test (stage 3b).
- **Unexplored:** a hind-leg loop through flybody's servos and its 22 claw neurons; hook and club neurons; campaniform sensilla.

D10. Drugs in the loop (stage 4)

- **Taken:** each knob acts on every network of a trial; movement disruption uses

log2((x + 0.01 mm)/(x_vehicle + 0.01 mm))	with the parent's 0.5 line, 0.05 mm as sensitivity.

- **Disclosed issue:** at full GABA-A block the reach depth is negative, so that condition's index averages the two proboscis terms. **Revisit:** a registered rule for negative depths.

D11. Checking against real flies (stage 5)

- **Taken:** a search protocol registered before any search; every candidate classified.
- **Unexplored:** neuron-specific simulations of the identified-neuron studies (DSOG1, 13A/13B) instead of class-wide knobs; a satiety state for the feeding model.

D12. Release (stage 6)

- **Taken:** a pipeline that reruns every later fly study as its own process, wired into `run_all.py` as a full-run-only `embodied` stage. **Why:** one command reproduces every number; separate processes keep the connectomes from accumulating in memory.

D13. Working overnight without the user (user: continue, commit, push, update docs)

- **Taken:** "push" read as merging finished work into master through a pull request, because GitHub keeps master only; every crossroads tonight is added here.

D14. The precision a chip needs, at the movement level (overnight)

- **Options:** leave stage 4a's timing result as it stands; compute, as the parent study did at the neural level, the chip's change at the dose where movement first fails.
- **Taken:** the latter, as an exploratory analysis (both inputs had been seen), added to the pipeline as `stage4a_precision`. **Why:** the paper's assay-design message rests on the precision a chip needs at behavioural onset; this extends it to movement at no simulation cost.

D15. Confirming the cell the corrected measure selects (overnight)

- **Options:** leave stage 2d's alternative selection unconfirmed (as the audit registered); confirm it on fresh seeds.
- **Taken:** confirm it (registration `embodied_stage2d_confirm_corrected.json`, commit 8495b57), with the corrected measure throughout. **Why:** it is the one loose end the audit named, and it is cheap. **Not taken tonight:** a hind-leg closed loop through flybody's servos (D9), a volume-based size rule (D7), neuron-specific drug simulations (D11), each larger.

D16. Bringing the embodied fly into the technical report and a paper draft (overnight)

- **Taken:** Act VIII of `report/TECHNICAL_REPORT.md` (§27-§31, every number cited, the uncited-number budget unchanged) and `report/PAPER_DRAFT.md` (abstract to methods, every number cited). **Why:** the user asked for a paper and for the report to keep being retouched; the draft is a starting point for the team, not a submission.

D17. Testing the model against identified-neuron experiments (overnight)

- **Options:** a hind-leg closed loop (D9), a volume size rule (D7), simulating the identified neurons that stage 5's published studies manipulated (D11).
- **Taken:** the identified neurons (stage 5b, registration commit c20fa08): the left front leg's GABAergic 13A and 13B hemilineage neurons, silenced (outputs removed) or activated (150 Hz), in the grooming relay of stage 1f. **Why:** it turns stage 5's indirect comparison into a direct one with data already in hand, and it is affordable overnight. **Not possible:** Pool et al.'s DSOG1 neurons are not annotated in FlyWire. **Detail:** 73 of the 76 annotated 13B neurons of that neuromere are in the nerve-cord network used.
- **Revisit:** subsets of 13A/13B (the study manipulated sub-populations), pulsed activation (the study's 70 ms pulses induced grooming-like rhythms), the femur-tibia angle as a readout ("legs locked in flexion").

D18. Reading the 13A/13B manipulations at the joint (overnight)

- **Options:** pulsed 13A activation (the study's grooming-inducing protocol; a leaky integrate-and-fire network has no rebound, so it would very likely fail by construction); a hind-leg closed loop (larger); the joint readout the study actually used.
- **Taken:** the joint readout (stage 5c, registration commit 04a2387, fresh seeds): femur-tibia angle under the same arms as stage 5b. **Why:** stage 5b's mismatch could be a readout mismatch (reach versus joint posture); this tests that directly and cheaply.
- **Process note:** a test edit for this step did not apply and a piped test run hid it; fixed in commit fd51761. Test runs before commits are no longer piped.

D19. Closing the loop on a well-traced leg (overnight)

- **Options:** stop stage 3 at the front-leg limit; close the loop on the right hind leg (25 claw neurons) through flybody's servoed tibia; build a muscle-driven hind leg (no such model at hand).
- **Taken:** the servoed hind leg (stage 3c, registration commit 22b082e): the walking command holds the right hind tibia extended (its extensor pool is tonically active, stage 2a), a 20 ms torque pushes it, and the return is timed under the resistance and assistance assignments and a sham, as in stage 3b. **Why:** it tests whether the front leg's failure was the thin sensor tracing, with the one body that has the leg. **Cost:** feedback changes a commanded angle, not muscle force; a servo's own stiffness sets much of the return.
- **Revisit:** a muscle-driven hind leg (if FlyMimic or another model adds one); the walking command's other legs.

D20. Testing stage 3c's exploratory difference on its own (overnight)

- **Options:** report stage 3c's resistance-against-sham difference as exploratory and stop; test it alone on fresh seeds.
- **Taken:** test it (stage 3d, registration commit 8ea194a): resistance against sham, paired by seed, 32 fresh seeds, a one-sided sign test and a 5 ms margin. **Why:** the difference was the one sign that the push-following part of the feedback acts on a well-traced leg, and it was seen in a noisy, unstable regime, so only a registered test on new seeds can say whether it is real.
- **Outcome:** 0 of 2 (stage 3 stays a documented negative). **Revisit:** a muscle-driven hind leg.

D21. After stage 3: replicate the headline before anything new (overnight)

- **Options:** new science that needs a model not at hand (a muscle-driven hind leg, neuron volumes for the size rule); 13A/13B subsets or pulsed activation (a leaky integrate-and-fire network has no rebound); paper and slide writing only; a registered replication of stage 4a, the paper's headline, which ran on one set of 8 seeds.
- **Taken:** the replication (stage 4b, registration commit b23f6a3): stage 4a's movements on 16 fresh seeds, against the chip detections of the parent design's own fresh-seed replication, with each movement left out in turn. **Why:** it is the result a reviewer will press first, it is cheap (about 15 minutes), and the leave-one-out check names the assumption most likely to make it wrong (that one movement carries the disruption index).
- **Outcome:** 3 of 4 (registration commit b23f6a3). The chip is late in every family on the fresh seeds; without the bitter-gated proboscis GABA-A potentiation ties the chip.
- **Revisit:** a different impairment line for movement (the 0.05 mm sensitivity is reported); a finer dose grid for GABA-A potentiation and excitability block, where the margin is one step.

D22. Resolving stage 4b's one-step margins (overnight)

- **Options:** leave the two thin margins reported; a finer dose grid inside them, as the motor-level tie-break did; a different movement impairment line (a new rule chosen after the results, so not taken).
- **Taken:** the finer grid (stage 4c, registration commit f387845): GABA-A potentiation at gaba_scale 1.6-1.9 and excitability block at threshold_scale 1.24-1.36 and 1.44-1.56, with stage 4b's movement seeds and both of the parent's chip seed sets, the coarse levels rerun as a reproduction check. **Why:** stage 4b showed that in two families the headline rests on a single step of the dose grid, and a finer grid is the direct way to ask what happens inside it.

- **Outcome:** 3 of 4 (registration commit f387845). Excitability block resolves chip later against both chip runs; GABA-A potentiation stays chip later with the full index, and tied without the bitter-gated proboscis.
- **Revisit:** a fourth movement (for example a walking readout, once one exists) would tell whether GABA-A potentiation's lateness is specific to feeding.

D23. Stopping new science for the night (overnight)

- **Options:** further stages (each remaining branch needs a model not at hand or is expected to fail by construction); documents for the team.
- **Taken:** a slide outline for the deck (`report/SLIDES_OUTLINE.md`, every number cited), the README's document table brought up to date, and a morning summary in `PROGRESS.md`. **Why:** the headline has now been tested as far as the current bodies allow (stages 4a-4c), and the next scientific choices (D7 neuron volumes, a muscle-driven hind leg, a walking readout for drugs) are the team's to make.

D24. The volume size rule (user)

- **Options (D7's unexplored branch):** neuron volumes from a future release, or MANC volumes through the male-CNS-to-MANC matching; found instead: the male CNS v1.0 neuPrint neuron table (same bucket, same CC-BY release) has a voxel count per body.
- **Taken:** fetch that table whole (4.65 GB, source `malecns_sizes`), checked against the bucket's CRC32C, because the bucket publishes no MD5 for it **(user: full file, not a column-only range read)**; rerun stage 2h with the published rule (stage 2i, registration commit d03ee27) on fresh draws. A numpy CRC32C was written rather than adding a dependency.
- **Outcome:** 6 of 7 (registration commit d03ee27): the rhythm needs the circuit and promotor and remotor keep a reliable, non-antiphase offset; flexor-extensor alternation fails.
- **Revisit:** normalizing by the median over the whole nerve cord rather than over the network (stated choice: over the network, as for the synapse-count stand-in).

D25. Checking that stage 2i comes from the neurons' own volumes

- **Options:** report stage 2i as it stands; name the assumption most likely to make it wrong and test it (the project rule for results that look good).
- **Taken:** stage 2j (registration commit 1d1d644), fresh draws: the volume rule replicated, volumes shuffled across neurons (a fresh permutation per draw), and normalization by the nerve cord's median instead of the network's. **Why:** if shuffled volumes do as well, the result is about the spread of sizes, not the identified neurons; the normalization was a stated choice the published description does not fix.
- **Outcome:** 2 of 4 (registration commit 1d1d644). Shuffled volumes give no rhythm; the cord-median normalization changes nothing; the rhythm and the circuit's necessity replicate, but the promotor-remotor offset does not reach the registered reliability, so it is not claimed.
- **Revisit:** a registered test of the offset alone with more draws, if it matters for the paper.

D26. Walking as a drug readout (user)

- **Options (from D22's revisit):** a fourth movement for the drug design; the only walking readout at hand is stage 2i's fictive front-leg rhythm (rate model, volume sizes, no body).
- **Taken:** stage 4d (registration commit 73ffde6), fresh draws: the drug knobs applied to the walking network (weights scaled by presynaptic class, as in the spiking model; the excitability knob scales the activation threshold), each draw's DNg100 input found under vehicle and held fixed, onsets on stage 4c's combined dose series. **Why:** it asks whether GABA-A potentiation's lateness is specific to feeding and whether the headline holds for a rhythm.

- **Stated choice:** the threshold knob maps to the rate model's theta; fixed inputs, so a drug changes the response, not the stimulus.
- **Outcome:** 2 of 4 (registration commit 73ffde6). GABA-A and GluCl block break walking first; acetylcholine block is chip-first; GABA-A potentiation and the threshold knob leave walking intact; with walking as a fourth movement the chip is still late in every family.
- **Revisit:** a readout that respects the bimodal draws (the fraction of rhythmic draws) registered as its own test; a threshold mapping that also shifts the drive.

D27. Retesting the acetylcholine exception with a readout that respects the draws (user)

- **Options:** keep stage 4d's median readout; a fraction-of-rhythmic-draws readout; a per-draw paired analysis.
- **Taken:** the fraction readout on fresh draws (stage 4e, registration commit 180476f). **Why:** stage 4d's draws are bimodal, the median sits between the modes, and the exception entered the abstract; the fraction is the plainest readout of a bimodal outcome and uses the rhythm line registered since stage 2g.
- **Outcome:** 2 of 4 (registration commit 180476f). GABA-A and GluCl block break walking first; the acetylcholine exception does not replicate: mild block raises the share of rhythmic draws, and the two-sided index counts that as a change before the chip. The exception is withdrawn from the abstract, discussion, README and slides.
- **Revisit:** a one-sided walking index (only loss of rhythm counts) would be a new rule chosen after seeing this result; register it only with a reason that does not depend on the outcome.

D28. The paper and new research move to a private repository (user)

- **Options:** keep everything public; make the repository private (would break the competition's links and the hosted demo, and erase stars and watchers); rewrite public history (destructive, and GitHub keeps every pull request's commits regardless); keep new work private from here on.
- **Taken:** new work private from here on. The public repository `Vodolay/AI4S` keeps the hackathon deliverables and everything already published (up to pull request 26). A private repository `Vodolay/AI4S-paper` mirrors public master and adds a `paper-draft` branch holding the paper draft, its outline, the slide outline (D23) and any new research toward the paper; those three documents were removed from public master. **Why:** the user wants the preprint and the research leading to it kept private until submission; what was already public stays so, and its history timestamps the work.
- **Revisit:** when the paper is submitted, `paper-draft` merges into public master.

D29. Recovering the ToxCast components that have no level-0 rows

(Numbered D55 in the project's private research log, where it was first taken.)

- **Found:** the invitrodb v4.3 dump's level-0 table lacks 8 of the 17 acute MEA components, among them firing rate, spike number and burst number; level 3 still has their rows.
- **Options:** use only the 9 components with level 0 (no firing rate); ask the EPA for the missing rows; recover the wells from the dump's id allocation.
- **Taken:** recovery (`neuromps.etl.toxcast_recover`). Level-0 ids come in contiguous runs of equal length per component and batch, so a component's run start is predicted from the reference component's; rval is recovered from level 3 exactly (rval = bval + resp x (pval - bval) / 100). Wells are assigned by position in the reference run; for all 7 missing electrophysiology components concentrations agree on every row and each date's stored baseline is reproduced from the assigned wells on every date. Four present components have their runs in another order (both checks fail on them), so present components keep their own level-0 wells.

- **Assumption recorded:** positions within a date are not checked by the baseline, so the physical-plate assignment of a recovered well rests on the run order alone.
- **Revisit at:** `data/interim/toxcast_mea/recovery_checks.json`.

D30. Testing the twin's dose response against real concentration-response wells

- **Found:** §33 showed the twin's knobs get the direction of the EPA acute screen's three positive controls right, but each is run at one saturating concentration, so §34 recorded the open question: nothing there tests the twin's dose response. The screen also runs about 500 compounds over seven concentrations each, with a viability readout on the same plates.
- **Options:** leave the dose response untested (the deck's closing slide already lists it as a next step); match knob levels to concentrations by potency (needs an IC50 per compound and per platform, which §14/HD8 showed does not transfer); match by rank and score only what a rank match can carry.
- **Taken:** the rank match, registered in `preregistration/twin_toxcast_dose_response.json` before any treated-well value was computed, with the compound list and one class label per compound fixed in the registration from published primary-target annotation. Scores are a Spearman rank correlation of the twin's matched curve against the real one per compound and readout, which readout moves first, and two nulls the twin must beat: the compound's own real firing-rate curve (the §33 follow-firing-rate rule, extended to curves) and the twin's curve under a knob the compound is not labelled with. **Why:** the knob scales and the concentration axis share no unit, and every registered score is invariant to a monotone map between them, so a rank match supplies exactly what the scores can use.
- **Registered before looking:** firing rate counts as already seen (a private analysis looked at it) and is excluded from the shape tests; bins are retained only up to the first that the screen's own viability readout flags, so that a collapse at the top concentration cannot be read as pharmacology; a compound needs four retained bins to be scored.
- **Assumption recorded:** the rank match assumes the screen's concentration range covers a comparable span of the mechanism to the knob grid. Checked by rescoring with the grid's two strongest levels dropped (`sensitivity_truncated_grid`).
- **Revisit at:** `outputs/metrics.json` key `twin_toxcast_dose_response`, report §35.

D31. Recalibrating the twin's inverse model: more simulations, then an ensemble

- **Found:** the inverse model (one masked autoregressive flow, 2,700 training simulations, `twin_hd14_inference`) fails the rank-uniformity check for `gabaa_scale` and `inter_in_degree`, so the slide withholds both.
- **Options:** keep withholding them; correct the check for five parameters (Bonferroni would pass both, which lowers the bar rather than fixes the model); retrain the same flow on more simulations; average several flows; change the flow family.
- **Taken:** a registered test of the two remedies that do not touch the bar: about 15,000 fresh simulations (5x; truncated at 05:00 EDT on 2026-10-28 by a prefix rule fixed in advance) for the same flow, and an equal-weight ensemble of five flows on the same pool, both judged on 1,000 fresh held-out simulations by the original per-parameter test at 0.05, plus coverage, contraction against the old model on the same simulations, and a check near the real recordings. The flow family is not changed, so budget and averaging are the only differences. **Why:** a slide that withholds two of five parameters should say whether that is a limit of the budget, of the estimator, or of what the summaries carry.
- **Registered:** `preregistration/twin_inference_recalibration.json` (seven predictions and the assumption check), committed before any new simulation.

- **Outcome:** 2 of 7 (registration commit 0bc29aa, amended before any simulation in 176b36e). 51 of 60 training chunks finished before the cut. On fresh simulations the round-1 model fails all five parameters, not two; the ensemble keeps its sharpness and passes four of five on the 300-run test, but fails `inter_in_degree`, stays over-confident by coverage for four parameters, and takes about 2 s per posterior. Budget alone fixes neither failing parameter. Under the registered rule the slide quotes `threshold_scale` only. Report §16, change-log entry 45.
- **Revisit at:** `neuromps.twin.recalibration`; `twin_hd14b_recalibration` in `outputs/metrics.json`. Next branches, each to be registered with fresh seeds: a different flow family (neural spline flow) on the same pool, which would separate estimator from summaries; or summaries that carry connectivity more directly (the failure that neither remedy moves).

D32. How noisy an untreated human iPSC chip is, against a rat chip from the same lab (user)

- **Found:** the practical-value result (§32) gives the wells an assay needs for rat cortical chips only. The natural human analogue, untreated wells recorded before and after a treatment step, exists in Kapucu 2022, but only 4 human and 6 rat such wells pass QC.
- **Options:** drop the human question; search for another licensed human iPSC set with re-recorded untreated wells; measure the noise inside one untreated recording, by comparing its two halves, on the lab's matched 48-well baselines (human DIV29, rat DIV22).
- **Taken:** the split-half comparison, registered in `preregistration/hpsc_split_half.json` before any half-recording feature was computed: three predictions (human firing rate noisier; noisier across features; replicates on the lab's 12-well developmental plates), and the wells per arm a 20% or 50% fall would need, reported as a lower bound. **Why:** it is the only comparison with enough wells in both species, and the same lab, protocol and extractor make the species the main difference.
- **Assumption recorded:** a species difference could be a firing-rate difference, because fewer spikes give noisier rates; checked on wells whose firing rates overlap between species.
- **Limit stated in advance:** split-half noise leaves out handling, compound addition and day-to-day variation, so the wells it implies are a lower bound, and are not compared numerically with the ToxCast vehicle-well numbers.
- **Revisit at:** `outputs/metrics.json` key `hpsc_split_half`.

D33. The feature definitions, recorded retroactively, and what they cost

- **What was decided, in Phase 1 (2026-09-16, `dd3d59b`):** an electrode is active at 5 or more spikes per minute; a burst is a run of at least 5 spikes with every inter-spike interval at most 100 ms; burst rate is bursts per minute per active electrode; the deposits' noisy electrodes are flagged in each record's QC block and not dropped; synchrony is STTC with a 50 ms window. All live as the defaults of `FeatureParams` in `src/neuromps/features/mea.py`, the one extractor the ETL and the twin both call.
- **What was not recorded at the time:** the rationale. The lab notebook's Phase-1 entry lists the values, and gives a source only for the activity threshold; the burst rule, the burst-rate denominator and keeping noisy electrodes were chosen without a written reason. This entry records them after the fact.
- **The reason, stated now:** one fixed, stated definition applied identically to every source and to the twin, so that a feature means the same thing in every record and in every simulation; the values follow the common fixed-threshold convention. The activity threshold is the one the EPA acute MEA assay states (its assay documents, e.g. `data/raw/toxcast_mea/assay_docs/AEID2466_CCTE_Shafer_MEA_acute_burst_p`

ercentage_mean.pdf, §2.5). The 50 ms STTC window is the one the Kapucu 2022 depositors state. Whether the burst rule matches a vendor's (Axion AxIS) default is **unverified**: no document in data/raw or the repository states it, so we do not claim it. Keeping noisy electrodes follows the standard's rule that upstream exclusions are preserved and flagged rather than silently dropped (schema/STANDARD.md §5).

- **What the choices cost (post hoc evidence):** src/neuromps/models/extractor_ladder.py reruns the extractor under the Kapucu 2022 depositors' stated settings one change at a time, against the meaRtools output they published. With their active threshold and their noisy-electrode lists, firing rate, active-electrode count and STTC reproduce theirs; most of the twofold burst-rate difference is the denominator, and it is an hPSC effect (report §3). The ladder closes a share of 0.9293 of the pooled burst-rate log-ratio gap. The detector difference (fixed rule against adaptive logISI) is named and not tested. This is post hoc: the result was seen before the code was written, and it is not registered.
- **Options:** keep the definitions (taken); switch the asset to meaRtools-like settings (rejected: it would rebuild every record and the twin's calibration targets for a gain on one source); or store features under named definition sets.
- **Taken:** keep the definitions unchanged; the extractor gains two options, off by default and named in the method record only when used (active_strict, burst_rate_denominator), so the asset, its method records and the twin are byte-identical.
- **Revisit at:** a future version could store each record's features under named definition sets (for example neuromps_v1 and meartools_like), selected per consumer, with the method record naming the set. outputs/metrics.json key extractor_agreement.settings_ladder.

D34. Testing whether the depositors' adaptive burst detector explains the rest of the extractor gap (user)

- **Found:** the post hoc settings ladder (D33, report §3) reproduced the depositors' firing rate, active-electrode count and STTC exactly and closed most of the burst-rate gap, but burst duration and spikes per burst did not move, and 10-15% of the hPSC burst-rate gap remained. The report attributed the rest to the burst detector (their logISI against our fixed 100 ms rule) without a test.
- **Options:** leave the attribution as an inference; implement logISI and test it.
- **Taken:** implement logISI (Pasquale et al. 2010) as an option of the single extractor, off by default, and register four predictions before running it on the Kapucu recordings (preregistration/extractor_logisi.json): burst duration and spikes per burst close, the hPSC burst rate closes, and the rat plate does not get worse. **Why:** a measured attribution answers "why trust your extractor" better than an inferred one, and the registration fixes in advance what would count as the detector explaining the gap.
- **Assumption recorded:** our implementation is close enough to meaRtools' that a failure speaks about the detector, not the implementation; the per-electrode thresholds and the fallback share are reported so a reader can judge.
- **Outcome:** 1 of 4 (registration commit e11bc05, implementation section completed before the run in 9570a9f). L1 and L2 fail on rank agreement with their ratios in band (burst duration 0.9611, rho 0.8347; spikes per burst 0.9986, rho 0.9467); L3 is uninterpretable (fallback on a share of 0.515 of the hPSC electrodes; both ratios overshoot); L4 passes. The detector does not explain the rest of the gap as registered, or our implementation differs; the attribution stays open. Report §3, change-log entry 48.
- **Revisit at:** outputs/metrics.json key extractor_agreement.logisi.

D35. Separating settling from steady noise in untreated human iPSC wells (user)

- **Found:** D32 (report §36) measured human iPSC firing rate about twice as noisy as rat between the halves of one untreated recording, and noted that human wells fall more within the recording; part of that spread may be wells settling at different speeds, not steady noise.
- **Options:** state the ambiguity and stop; separate the two within the same recordings.
- **Taken:** six windows per 30-minute baseline, a straight-line trend per well on a fixed electrode set, and three registered predictions (`preregistration/hpsc_settling.json`): human wells settle more unevenly; human wells are also noisier around their own trend; in human wells the trend carries most of the split-half spread. **Why:** for assay design the two call for different fixes (a longer settling period before dosing against more wells), so which one dominates matters.
- **Assumption recorded:** a straight line describes the settling; checked with a quadratic term.
- **Outcome:** 3 of 3 (registration commit e743c4c; code and tests committed before the first real-data run in bc6a56b). Human slopes spread 3.65 times as widely as rat; the trend carries a share of 0.996 of the human split-half variance. The linearity check fails for human wells (a fall and partial recovery), so T1 and T3 carry that caveat, and T2's residual SD is mostly curvature, not steady noise. The T3 threshold turned out to sit below what a line absorbs from pure noise (about 0.77); recorded, not changed. Reading: D32's human spread is mostly uneven settling; a longer settling period is the first fix to try, but five more minutes would not close the gap (post hoc), and the 30-minute recordings do not show how long it must be. Report §36, change-log entry 49.
- **Revisit at:** `outputs/metrics.json` key `hpsc_settling`. Next: longer untreated human recordings, to find where the fall ends.

D36. Testing whether early network bursting predicts how a human iPSC well settles (user)

- **Found:** an exploratory look at D35's human wells (post hoc, 24 wells, one plate) found two settling behaviours, wells that keep falling and wells that recover, and the network-burst rate in the first five minutes predicted which (Spearman -0.73), while the early firing rate did not; rat wells showed no such relation.
- **Options:** report the exploratory result as it stands; replicate it on independent recordings first.
- **Taken:** a registered replication on the lab's 12-well developmental plates (`preregistration/hpsc_network_state.json`), which D32 already uses as its replication pool: three predictions (the relation holds in human recordings; not in rat; network state predicts better than firing rate), with intervals from resampling whole wells. **Why:** a post hoc finding on one plate is a hypothesis; if it replicates, a lab could cut the wells it needs by stratifying human wells by baseline network state.
- **Assumption recorded:** the relation is not an age effect; checked with a partial correlation on culture age.
- **Outcome:** 1 of 3 (registration commit f669016; code and tests committed before the real-data run in d7ba7da). N1 fails: in 144 human developmental recordings rho(nb0, late_b) is −0.06, interval −0.23 to 0.09; N2 passes (rat rho −0.06, no relation in either species); N3 fails. The age flag is not raised (human partial rho −0.39), but the per-DIV median (−0.06) does not agree with it. The post hoc baseline result recomputes as −0.73 and stays a hypothesis tied to one plate. Reading: stratifying human wells by baseline network bursting is not supported by the replication. Report §36, change-log entry 50.
- **Revisit at:** `outputs/metrics.json` key `hpsc_network_state`. Next, if pursued: 30-minute untreated recordings of more human plates at matched ages.

D37. Storing each feature under named definition sets (engineering)

- **Found:** D33 kept one fixed set of feature definitions and left a future option open: store each record's features under named definition sets. The post hoc settings ladder (report §3, `extractor_agreement.settings_ladder`) showed that with the Kapucu 2022 depositors' stated settings our extractor reproduces their firing rate, active-electrode count and STTC; the registered logISI test (D34, `extractor_agreement.logisi`) passed 1 of 4 predictions with the attribution open. A lab reading our numbers beside theirs had no way to see which of our features translate into theirs, and which do not.
- **Options:** keep one fixed definition only, and leave conversion to the reader (rejected: the evidence that tells a reader where conversion works would stay in a metrics file); recompute everything per lab on demand (rejected: every consumer would rerun the ETL, and nothing would pin what a named lab definition means); named definition sets plus a sidecar table (taken).
- **Taken:** a registry, `src/neuromps/features/definitions.py`, of three frozen sets, each a named bundle of options `features/mea.py` already has plus whether a deposit's noisy electrodes are dropped: `neuromps` (reference, the defaults the asset and the twin use), `kapucu2022` (validated, the ladder's S3) and `kapucu2022_logisi` (experimental, S4, used for no claim). The ladder and the logISI study now take their S3 and S4 from the registry; their definitions are equal to the old ones, so their outputs do not change. A sidecar, `data/processed/features_by_definition.parquet`, holds every spike-time record under every set, re-read with the ETL modules' own loaders; its `neuromps` rows must equal the asset exactly. A translation table (`outputs/tables/feature_definitions.md`) gives, per feature, the ratio and rank agreement between our numbers and each Kapucu set on the recordings where the depositors' own output exists, and a verdict: same number, scale factor or no conversion. **Why:** additive, so no record, method block, twin target or earlier metric moves; one extractor still computes everything; and a consumer can see, per feature, whether a translation is supported by evidence before converting.
- **Not a registered study:** it computes no new comparison with the depositors. The translation re-expresses the ladder's (post hoc) and the logISI test's (registered) evidence per pair of definition sets.
- **Assumption recorded:** a pooled ratio between two definitions is a property of the recordings it was measured on, not of the definitions. The translation table reports the ratio per plate beside the pooled one, so a factor that differs between the human and rat plates is visible, and it is checked only on the Kapucu 2022 developmental plates.
- **Revisit at:** `outputs/metrics.json` key `feature_definitions`. Next: a set for another laboratory only when that laboratory's own output exists on spike files we hold (the condition for a `validated` status); promoting `kapucu2022_logisi` only after its attribution is resolved (D34's next step, reading the depositors' scripts).

D38. Is the twin's distance from the real recordings the prior's range or misspecification?

- **Found:** D31 (report §16) found the 113 real two-chamber recordings far from every twin simulation and calibration failing near them. Before registering, a diagnostic on HD14's pool (declared as seen) located the gap: on HD14's own scale the median distance from a real recording to its nearest simulation is 2.43 against 0.409 between neighbouring simulations; `log_rate_ratio` carries a third of it (real recordings run faster in compartment 0 than any simulation), `lag_ms` a sixth and the active-electrode counts a quarter; HD14's posterior, shown the real recordings, puts most of its raw mass above the `threshold_scale` bound (72% of recordings) and some below the `log10_forward_ratio` bound (27%), while `inter_in_degree` and `gabaa_scale` stay inside.

- **Options:** report the gap as misspecification without a test; widen every bound; widen or re-centre only the bounds the real recordings push against; change the model (a parameter for compartment-specific excitability, conduction delay, electrode coverage).
- **Taken:** a registered test of the third option, because it is the one that can separate the two readings at fixed model: `log10_forward_ratio` re-centred to -1.3..1.3, `ampa_scale` to 0.4..1.8, `threshold_scale` to 0.85..1.35, the other two unchanged; at most 3,000 new simulations (600 held out, simulated first) with a cut-off at 06:00 New York time on 2026-10-29; distances on one fixed scale (HD14's pool), at equal pool sizes; R1 (reach: the median real-to-nearest distance at most halves), R2 (the old region keeps a neighbour for at least half of HD14's held-out simulations), R3 (one HD14-recipe flow on the new pool passes the KS rank test for `threshold_scale` on the 200 held-out simulations nearest the real recordings). **Why:** whether the twin's inverse model can ever be read on real recordings depends on which of the two it is, and only widening where the posterior pointed tells them apart without changing the model.
- **Assumption recorded:** HD14's cached summaries are what today's code produces, although the summary-code hash moved after two off-by-default extractor options were added (checked by re-simulating six rows exactly); and an R1 pass is not an artifact of HD14's scale (checked on two other scales). A misspecification reading is expected in any case for the part of the gap carried by `lag_ms` and by electrode counts above the layout's 48.
- **Registered:** `preregistration/twin_prior_reach.json`, committed before any simulation from the new prior.
- **Outcome:** 2 of 3 (registration commit 3c2e43e at 11:32 EDT; the first simulation started at 11:33). All 3,000 new simulations finished well before the cut-off. R1 fails: the median real-to-nearest distance falls only from 2.43 to 2.211 (ratio 0.91), on every scale checked, and the same summaries carry what remains. R2 passes (share 0.8233), and R3 passes for `threshold_scale` near the real recordings (p = 0.2682) while `gabaa_scale` and `inter_in_degree` fail there. HD14's cache reproduces exactly under today's code. Reading, as registered: misspecification within the widened box, not the prior's range; the inverse model is not to be read on absolute real recordings. Report §16, change-log entry 53.
- **Revisit at:** `neuromps.twin.prior_reach`; `twin_prior_reach` in `outputs/metrics.json`. Next: a model change, registered with fresh seeds: a chamber-specific excitability parameter and a coupling-delay parameter, the two things the remaining summaries point to. The new flow still puts most raw mass above the new `ampa_scale` bound for 30% of recordings, so a still wider box for that knob is not ruled out.